

TABASCO TECH CONT LLC

Tabasco Legal Management System

Complete Architecture, Functional Modules & Reference Manual

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Developed By:	ARABINFOTEC
Target Platform:	PHP 7.4+ & MySQL (WAMP / Linux)
Core Libraries:	mPDF v8.2, FPDF, PhpSpreadsheet, PHPMailer

1. Application Overview

The **Tabasco Legal Management System** is a highly customized web platform designed to streamline legal proceedings, client collections, court schedules, and agency payouts for **TABASCO TECH CONT LLC** (UAE). The system digitizes legal operations, providing centralized transparency across external legal firms, independent debt collectors, and internal legal coordinators.

Key Business Capabilities:

- **Client & Case Onboarding:** Seamlessly register corporate clients, generate unique identifiers, and initiate legal cases.
- **Financial Auditing:** Dynamic tracking of total outstanding balances, Post-Dated Cheques (PDC), collections, court fees, and third-party commission distributions.
- **Collaborative Portals:** Restricted login pathways for third parties, external lawyers, and debt collectors to view cases assigned directly to them.
- **Document Repositories:** Categorized document management mapping templates and legal proof materials directly to client profiles or active files.
- **Activity Logging:** Every insert, edit, soft delete, and login is auditable, tracking before-and-after snapshots to ensure maximum data accountability.

2. Technology Stack

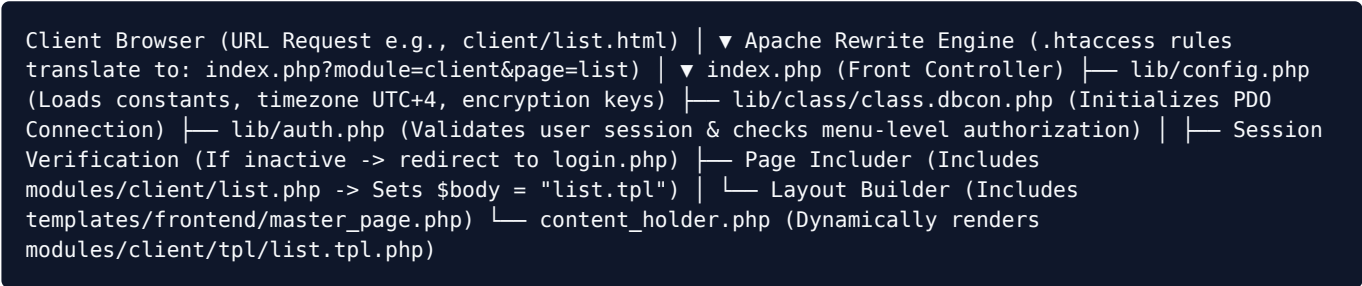
The software utilizes a lightweight, high-performance architecture relying on standard web components without heavy third-party framework overhead, ensuring rapid page loads and easy maintenance.

Layer	Technology	Operational Details
Backend Engine	PHP (Procedural + OOP)	Implements a lightweight, custom MVC-inspired request routing paradigm.
Database Layer	MySQL	Utilizes PDO (PHP Data Objects) with persistent connections and strict prepared statements.
Frontend Theme	Bootstrap 5 + jQuery	Fully responsive, customized layout supporting dark/light mode toggles.
PDF Generation	mPDF v8.2 + FPDF	Generates vouchers, receipts, and statement files dynamically.
Spreadsheet Engine	PhpSpreadsheet	Exports tabular list records and statement analytics directly to Excel format.
Mail Delivery	PHPMailer	SMTP-based mailing classes for automated notifications and PDF receipt mailings.

3. System Architecture & Routing

The application routing flows through a front controller pattern. Apache rewrite rules (.htaccess) intercept requests, rewriting them into parameter inputs resolved by `index.php`.

Request Lifecycle Diagram:



Clean URL Configuration:

Clean URL Pattern	Standard Query Map
<code>{module}.html</code>	<code>index.php?module={module}</code>
<code>{module}/{page}.html</code>	<code>index.php?module={module}&page={page}</code>
<code>{module}/{page}/{action}/{param1}.html</code>	<code>...&action={action}&param1={param1}</code>

4. Security & Authentication Framework

Securing case details and financial collection audits is crucial. The platform applies multiple defensive layers:

- **SQL Injection Prevention:** No variables are interpolated directly into SQL blocks. Queries run through PDO parameter binding (`?` or `:name`) utilizing the wrapper class.
- **Password Storage:** Passwords are encrypted using high-entropy `**AES-256-CBC**` encryption keys stored inside config variables, decrypted on request using `decryptPassword()`.
- **Cross-Site Request Forgery (CSRF):** Form submissions check dynamic session-based tokens generated via `bin2hex(random_bytes(32))` and verified through `class.crsftoken.php`.
- **Soft Delete Architecture:** Toggling records sets the status flag to 'D'. Data is retained in database storage, but default SELECT queries filter them out unless explicitly required.

5. Core Functional Modules

The system core is categorized into 14 distinct functional modules that interact dynamically to manage legal workflows:

5.1 Dashboard Module

Location: modules/dashboard/ | **Primary View:** panel.tpl.php

Renders user-specific items on login. Evaluates user permissions to compile active reminders and logs.

- Fetches date-bound reminders using `LegalTask_reminders::get_taskReminders()`.
- Pops up calendar case notifications via `Casenotification::get_notifications()`.

5.2 Client Management

Location: modules/client/ | **Class:** Clients | **Table:** legal_client

Maintains client corporate details, upload categories, and financial outstanding balances.

Core Operations	Database & Code Hook
Registration & Editing	<code>Manage_Client_information()</code> handles inserting and updating master profiles.
Code Auto-generation	Appends zero-padded database IDs onto the prefix <code>TABL/CL/</code> .
Collateral Files	Stores visiting card and client file references in <code>legal_document</code> .
Outstanding Updates	Monitored via <code>Update_Cheque_OutStanding()</code> . Modifying these fields triggers financial audit logs.

5.3 Active Legal Management

Location: modules/activelegal/ | **Class:** ActiveLegal | **Table:** legal_activelegal

Registers main case profiles. Links client information to assigned agencies. Tracks claims, payments, and agency shifting history.

- **Agency Shifting:** Handled by `shift_active_legal()`. When a client case shifts from a third party to a new lawyer, the old relationship closes, and shifts are logged in `legal_shift_active_legal`.
- **Commission Setup:** Handled by `save_commission()`. Configures default payout percentages per agency.

5.4 Case Management

Location: modules/case/ | **Class:** LegalCase | **Primary Table:** legal_case

Tracks legal case actions and hearings. Integrates the **Case Roots** system to handle nested sub-stages.

Case Root Flow (Multi-Stage Court Tracking): Legal Case (Master ID) — Case Root (Identifies current trial phase e.g., First Instance, Appeal) — Court Details & Category Definitions — Root Actions (Log milestone events, PDF judgments, UAE Pass details)

Method Signature	Operational Purpose
saveCase(\$data, \$id)	Saves basic case structures (case number, court, category, plaintiff, defender).
saveRoots(\$data, \$id)	Creates or updates stage records for nested court level processes.
syncCaseRelations(\$c_id, \$r_ids)	Synchronizes bidirectional relations for connected legal files.

5.5 & 5.6 Collections & Expense Management

Classes: Collection & Expense | **Tables:** legal_collections & legal_expense

Tracks cash inflows (collections) and cash outflows (expenses). Serves as critical input for computing outstanding balances.

- **Collections:** Records amounts collected via cash, cheque, or bank transfer, referencing fee types and optional document attachments. Recalculates remaining claim totals.
- **Expenses:** Records court filing fees, processing fees, translation costs, and lawyer fees related to a case.

5.7 & 5.8 Commission Payouts & PDF Vouchers

Classes: LegalCollectionCommission & LegalCommissionVoucher | **Tables:** legal_collection_commission & legal_commission_voucher

Processes calculated commissions to generate official payment vouchers, renders voucher documents as PDFs, and updates transaction states.

- **Calculation:** Triggers on collection entries, saving commission parameters into `legal_collection_commission`.
Formula: $\text{Payable} = (\text{Collected Amount} * \text{Commission \%}) / 100$.
- **PDF Generation:** The mPDF utility translates HTML layouts into vouchers, assigning unique voucher numbers, archiving PDFs in `uploads/expenses/`, and mapping records in `legal_commission_voucher`.

5.9 Document Management

Class: processDocument | **Tables:** legal_document, legal_document_type

Attaches uploaded files to specific entities using parent codes ('C' for Client, 'AL' for Active Legal, etc.). Validates file types (JPG, PNG, PDF) and enforces a 1MB file size limit.

5.10 Reports Module

Folder: modules/reports/

Compiles multi-dimensional operational metrics. Includes data tables and export parameters for business intelligence.

Report Type	System URL Mapping	Description
Bad Debts Report	reports/bad_debts.html	Tracks outstanding client files flagged as bad debt.
Closed Legal Report	reports/closed_legal_report.html	Displays closed, archived, and resolved case records.
Total Legal Statement	reports/total_legal_report.html	Aggregates active case timelines and financial stats.

5.11 Tasks & Reminders

Class: LegalTask_reminders | **Table:** legal_task_reminders

Powers calendar tasks and reminders, integrating directly with dashboard displays. Helps users track court dates, document submission deadlines, and follow-ups.

5.12 Master Data Module

Folder: modules/master/

Maintains system-wide lookup tables for dropdown menus, configuration records, and system-wide attributes. Includes banks, courts, document types, case categories, fee types, and locations.

5.13 User Access Control & Permissions

Class: UsersClass | **Table:** users, usertype

Controls system access. Superadmins map granular permissions (View, Add, Edit, Delete, Print, Mail) to menus for normal staff, while agency users are restricted to assigned files.

5.14 Agency Management

Coordinates external workflows, allowing external entities (lawyers, firms, debt collectors) to log into restricted dashboard interfaces and update assigned files.

6. Primary Database Directory

Below is a consolidated directory of the primary database tables that drive the application layout and relationship structures:

Table Name	Operational Description
legal_client	Maintains primary details of registered clients.
legal_activelegal	Tracks active cases, linked agencies, claim values, and current status.
legal_case	Contains judicial details (case number, court, category, plaintiff, defender).
legal_case_roots	Maintains multi-stage court trial milestones (First Instance, Appeal, Supreme).
legal_collections	Logs financial collection inflows and links them to cases and client accounts.
legal_expense	Logs court fees, translation, lawyer payouts, and case expenses.
legal_collection_commission	Calculates payouts for third parties, legal firms, or collectors.
legal_commission_voucher	Logs generated commission vouchers and links them to saved PDFs.
users	Maintains user account credentials (encrypted passwords) and access profiles.
legal_activity_logs	Maintains audits of all data operations, including before-and-after snapshots.

7. Core Application Workflows (How It Works)

The Tabasco Legal Management System orchestrates data entry and financial operations through structured step-by-step workflows. Below is an operational walkthrough illustrating the lifecycle of clients, cases, collections, and commission payouts:

Step 1: Client Registration & Onboarding

The workflow begins by registering corporate or individual clients into the master database:

- Navigate to **Client Management** and click **Create Client**.
- Fill in client corporate details, marketing agent references, address records, and upload a digital copy of their visiting card (enforced under a 1MB limit).
- Upon saving, the database triggers `Get_Last_Client_ID()` to append a zero-padded sequential ID onto the default prefix, yielding a code like `TABL/CL/0042`.

Step 2: Opening an Active Legal File

An Active Legal file is the core container that bridges a client profile to an assigned legal agency or internal representative:

- Navigate to **Active Legal Management** and select **Add Active Legal**.
- Associate the file with a registered Client, select the legal Status, and assign the file to an agency (Third Party, External Legal Firm, Debt Collector, or internal team).
- Define the baseline financial metrics: Claim Amount, initial outstanding amounts (with/without cheques), and the default commission parameters.
- The system generates a tracking code like `TABL/AL/0018` and logs the transaction.

Step 3: Initializing Judicial Court Cases

Once an active legal file is opened, specific court case records are created to track lawsuits and hearings:

- Under the Active Legal record, choose **Initialize Court Case**.
- Input the official Case Number, select the Court branch, assign the presiding Lawyer, specify plaintiffs/defendants, and set the Case Mode (e.g. First Instance).
- The case inherits default parameters from the Active Legal file, but allows overriding outstanding balances.

Step 4: Scheduling Hearings & Logging Root Actions

Lawyers and legal coordinators record litigation progress on a daily basis:

- **Hearings:** Add scheduled hearing entries with trial feedback notes and specify the next hearing date. The system automatically elevates the latest hearing date onto the master case view.
- **Case Roots (Sub-Stages):** Create root containers representing major trial stages (e.g., appeal filings or executions).
- **Root Actions:** Inside each trial root, log chronological action milestones (such as application submissions or order releases), upload judgment PDFs, and attach UAE Pass verification reference codes.

Step 5: Recording Payment Collections

When an agency or legal coordinator recovers funds from a debtor, a collection entry is recorded:

- Navigate to the collection log within the case details, and click **Record Collection**.
- Specify the received amount, date, fees type, payment mode (cash, bank wire, cheque), and upload the deposit receipt.
- The system saves the record in `legal_collections` and dynamically triggers balance reconciliation, subtracting the collected sum from the client's outstanding totals.

Step 6: Automatic Commission Processing & Voucher Payouts

The software handles automated payouts for third-party debt collectors and legal firms:

- When a collection is saved, the system checks the active legal commission setup and automatically computes the payable share:
$$\text{Payable Commission} = (\text{Collection Amount} * \text{Commission \%}) / 100.$$
- This calculation is saved in the `legal_collection_commission` queue.
- Go to the **Commission Report** interface, select the calculated commissions, and click **Generate Payout Voucher**.
- The system instantiates the mPDF class to render a formal voucher document containing a breakdown of collections and payouts, saves the PDF as `Commission_Voucher_VC-XXX.pdf` under `uploads/expenses/`, creates a voucher record in `legal_commission_voucher`, and updates the payment status to **Printed**.
- Once the physical payment is processed, the status is toggled to **Paid** to finalize the transaction.

8. Deployment & Environment Setup

The software runs in local WAMP developer environments and on live Linux servers, utilizing standard configuration adjustments.

Environment Configuration (lib/config.php):

Toggle credentials by modifying database parameters and base URL schemas inside the config script:

```
// Database Settings
define("IP", "localhost");
define("DB", "demouser_talegal");
define("USER", "demouser_talegal");
define("DBPWD", "mevda5GM4SDD");

// Base URL Definitions
define("ROOT_DIR", "https://yourdomain.com/");
define("AJAX_ROOT_DIR", "https://yourdomain.com/");
```

Required PHP Extensions:

- `pdo_mysql` — Standard database abstraction routing.
- `openssl` — Powers AES-256-CBC password encryption/decryption.
- `mbstring` / `gd` — Render engines for mPDF layouts and images.

Directory Permissions (Write Access Required):

- `uploads/` — For visiting cards and root attachments.
- `files/` — For generated statement data sheets.
- `vendor/mpdf/mpdf/tmp/` — Temporary files and font caches for PDF generation.